ZIGBEE vs SUZI

Feature	Zigbee	Suzi
Frequency band	2.4GHz	Sub-GHz, including 800 MHz in Europe and 900 MHz in North America
Typical range	Suited for short- to medium-range IoT networks	Longer range; sub-GHz can reach 1 km or more in some deployments
Wall penetration	Can be affected by walls and other obstacles	Better penetration through walls and other physical obstacles
Mesh networking	Yes	Yes
Power consumption	Low power	Retains Zigbee's low-power approach
Network architecture	Zigbee mesh network	Uses the existing Zigbee network-layer routing and mesh approach
Spectrum	Shares the crowded 2.4GHz band with Wi-Fi, Bluetooth, and other devices	Uses less-crowded sub-GHz bands, depending on region
Data throughput	Higher than sub-GHz options in general	Lower throughput, suitable for low-bandwidth IoT applications
Large-area deployment	May require more routers/repeaters as distance increases	Better suited to large buildings, factories, utilities, and outdoor areas
Compatibility with existing Zigbee	Existing Zigbee network	Can work alongside 2.4GHz Zigbee through suitable bridge devices
Best suited for	Homes, rooms, lights, sensors, switches, and other general IoT devices	Long-range sensors, meters, garages, basements, factories, outdoor infrastructure, and large sites

architectural updates introduced in Zigbee 4.0, including enhanced security, commissioning, network reliability, and device management. Zigbee 4.0 maintains strict backward compatibility with older deployments while allowing manufacturers modular flexibility in choosing which new capabilities their hardware supports.

Security and interoperability

Zigbee 4.0 strengthens network access controls to block unauthorised access and replay attacks. By introducing advanced security provisions at the application support layer, it couples flexible cryptography with precise device authentication, key management, and network protection. Key features include Dynamic Link Key, Device Interview, Secured Channel, Restricted Mode, PAN ID Changes, Trust Centre Swap Out, and Frame Counter Synchronisation.

While backward compatible with Zigbee 3.0 and legacy Smart Energy solutions, mixed-frequency deployments introduce unique design challenges. Operating across multiple personal area networks (PANs) requires careful optimisation within the software stack and physical radio layer. Ultimately, reliable interoperability between 2.4GHz and sub-GHz nodes hinges on how effectively vendors implement the underlying multi-radio hardware architecture.

What this could mean for IoT deployments

- ***Broader deployment sectors.*** Suzi will make it possible to implement the IoT systems over large areas, including residential, warehouse, factory, utility, and outdoor infrastructures, where wireless communication would usually pose problems.
- ***Easier scalability.*** Existing Zigbee systems can be scaled by deploying Suzi modules and bridges where necessary to incorporate sub-GHz wireless connectivity.
- ***Less maintenance.*** The ability to communicate over long range, combined with Zigbee's low-power consumption benefits battery-operated sensors and meters deployed in hard-to-access areas.
- ***Improved wireless performance.*** Sub-GHz wireless operation proves useful in cases where obstacles such as walls and distances make wireless communication difficult.
- ***Less data throughput.*** Along with its long-range advantages, sub-GHz communication technology has slower data throughput, making Suzi well-suited for low-bandwidth IoT applications.
- ***Certified products.*** The new certification program makes it easier for manufacturers to certify their products for interoperability in a multi-vendor environment.

What does the certification opening mean?

Suzi itself is not a brand-new technology; it originated during the development of Zigbee 4.0. But what is new is that the Connectivity Standards Alliance (CSA) has officially launched the Suzi Product Certification Program. Previously, vendors could develop sub-GHz hardware around the specification, but they lacked a formal compliance pathway to validate their products within the broader Zigbee ecosystem.

This program introduces a standardised testing framework to verify specification compliance and multi-vendor interoperability. Passing certification grants hardware official listing in the CSA database, giving system integrators a clear, validated path to deploy these sub-GHz devices across challenging environments like large buildings, smart meters, factories, outdoor networks, and smart cities.

With the certification ecosystem active, the critical next step is for silicon vendors and device manufacturers to transition the Suzi specification into scaled commercial deployments. EFY

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